

How Effective are Enforcement Efforts Targeting Illegal Drugs?

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Research Question

- Specifically
 - Does over-the-counter (OTC) regulation on sales of ephedrine and pseudoephedrine have significant impact on methamphetamine production, market, consumption and drug related crime

Research Question Importance

- Why is this so important?
 - Drugs threat our social harmony
 - Pollution during production
 - We do not want drugs
 - We wonder how to reduce drug production, consumption and drug related crime
 - We wonder if enforcement efforts are effective

Introduction

- Methamphetamine
 - A very effective kind of drug
 - Extremely easy to manufacture
 - Requires ephedrine or pseudoephedrine as precursors
- Ephedrine/ Pseudoephedrine
 - Requires large facilities to manufacture
 - Available in drug stores as cold medicine

OTC Law

- Between 2004 and 2006, 35 states passed laws regulating the sale of OTC products that contain pseudoephedrine or ephedrine
 - Limit the sale of the products
 - Place the products out of customer's reach
 - Purchasers need to present ID
 - Retailers maintain a logbook

Identification Strategy

- Both the time period and geographical area affected by the enforcement are known and well-defined
- A difference-in-difference model can be implemented
- Assumption: there are not unmeasured state specific trends correlated with the enactment of the regulation

Data

- Methamphetamine Production
 - Cannot observe
- Methamphetamine Market
 - Cannot observe
- Methamphetamine Use
 - Cannot observe
- Drug Related Crime
 - Cannot observe

Methamphetamine Production

- The National Clandestine Lab Seizure System
 - Contains information on labs seized or discovered by law enforcement
 - Contains estimates of each site's production capacity
 - Use changes in number of detected labs to proxy changes in number of all labs

Methamphetamine Market

- Drug Enforcement Agency's System to Retrieve Information from Drug Evidence
 - Record the date, location, method of acquisition
 - Purity
 - Price

Methamphetamine Use

- Quest Diagnostics
 - Workplace drug tests
- Health Care Utilization Project
 - Identify hospital admissions (for any cause) that tested positive for amphetamine

Methamphetamine Use

- Previous studies use voluntary survey data
 - Able to randomly select sample
 - Can hardly reveal the truth since people have incentive to hide illegal activities
- This paper uses mandatory drug test data
 - Can reveal the truth within sample
 - Sample is restricted to workplace and hospital
 - Potential selection bias

Drug Related Crime

- Uniform Crime Reports
 - Arrest rates for the possession and sale of illegal drugs for each state and month

Event Study

$$Y_{st} = \sum_{j=-10}^9 \pi_j 1(\tau_{st} = j) + \pi_{10+} 1(\tau_{st} \geq 10) + \alpha_s + \gamma_t + \delta_s * t + \omega_s * t^2 + \epsilon_{st}$$

- Examine the timing of any change in each outcome
- Examine any “pre-trend” that would raise concerns about the identification strategy
- Trace out the dynamics of any policy effect

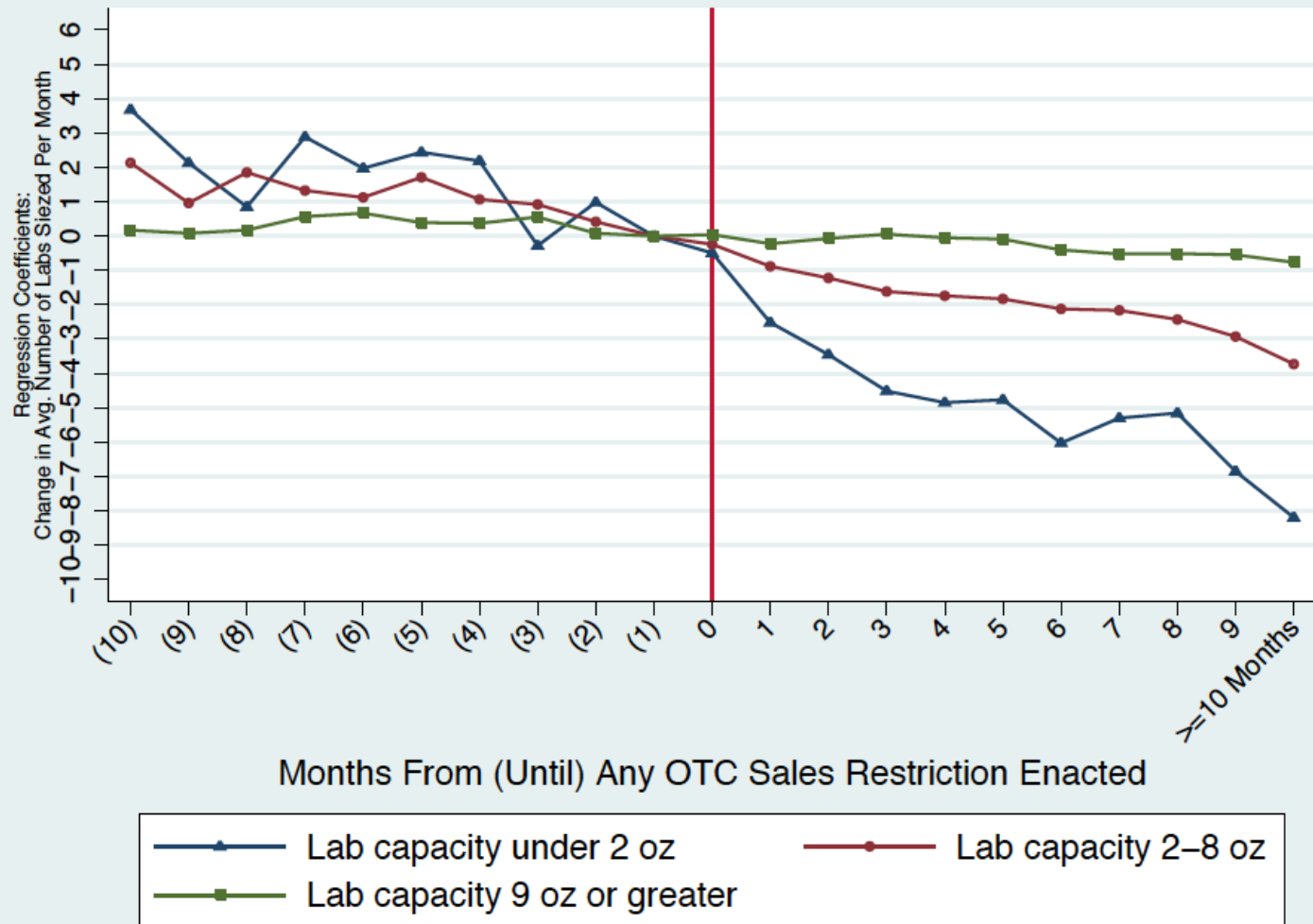
Model

- (1) $Y_{st} = \beta OTC_{st} + \alpha_s + \gamma_t + \epsilon_{st}$
- State fixed effect: α_s
- Time fixed effect: γ_t
- Cluster standard errors by state
- Extend the model by adding
 - (2) State specific linear time trends
 - (3) State specific quadratic time trends
 - (4) Covariates with possibility to be confounded with outcome
 - Unemployment rate
 - Number of households receiving food stamps
 - Average temperature
 - Average precipitation

Results

- Methamphetamine Production
 - Number of lab seizure significantly reduced
- Methamphetamine Market
 - No significant effect on nominal price or purity
- Methamphetamine Use
 - No significant effect on any of the proxies
- Drug Related Crime
 - No significant effect on arrest rates

Methamphetamine Production

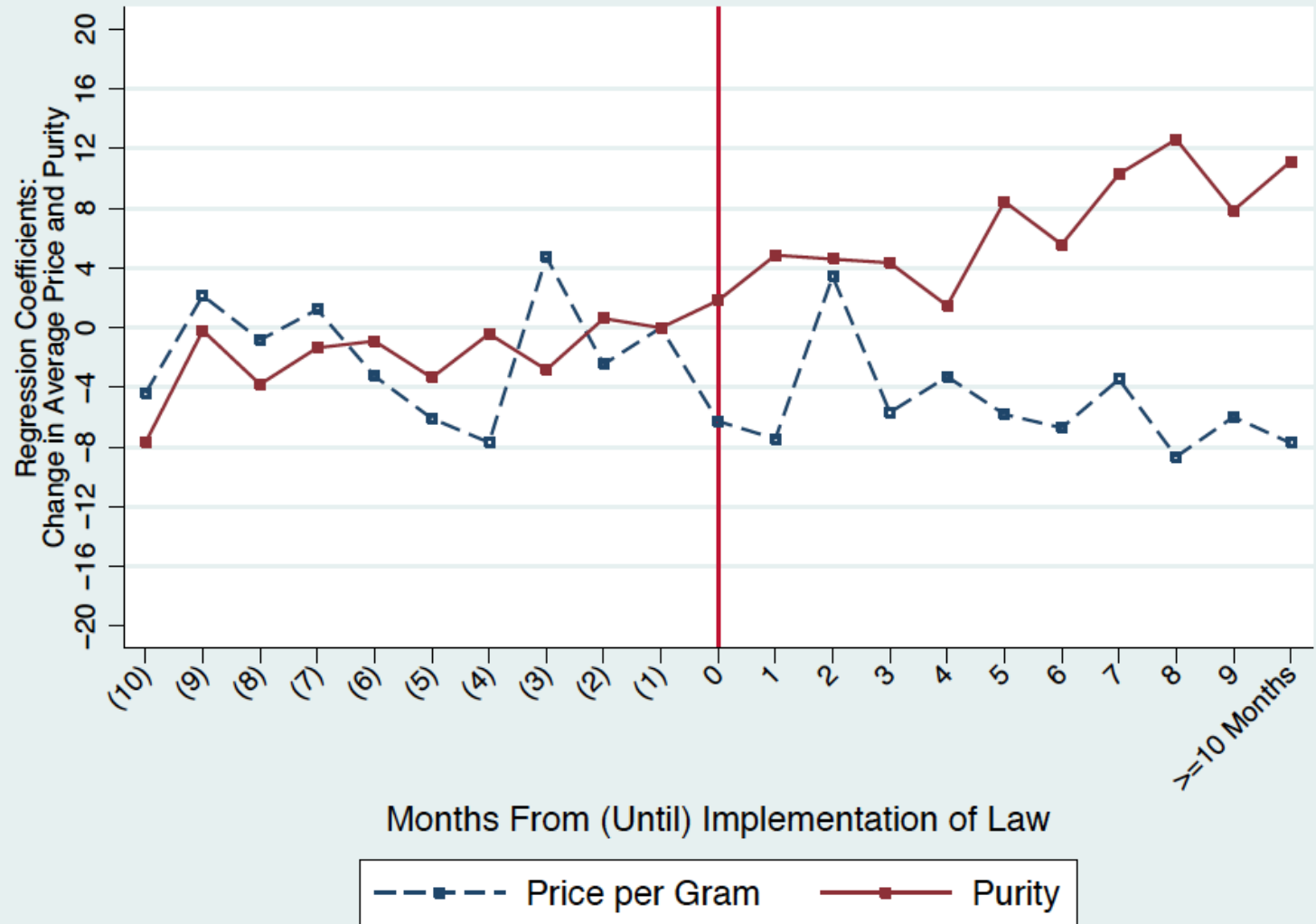


Methamphetamine Production

	<u>Number of Labs Seized</u>				<u>Lab Capacity Under 2 Ounces</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-6.71 (1.52)	-5.05 (1.44)	-7.02 (1.87)	-7.30 (1.93)	-4.08 (0.90)	-3.05 (0.89)	-4.73 (1.31)	-4.92 (1.36)
Mean Prior to OTC Restriction	13.94	13.94	13.94	14.46	9.59	9.59	9.59	9.93
Observations	3825	3825	3825	3675	3825	3825	3825	3675
Number of States	51	51	51	49	51	51	51	49

	<u>Lab Capacity 2 to 8 Ounces</u>				<u>Lab Capacity 9 Ounces or More</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-2.29 (0.55)	-1.76 (0.47)	-1.87 (0.56)	-1.94 (0.58)	-0.34 (0.22)	-0.24 (0.19)	-0.42 (0.21)	-0.44 (0.21)
Mean Prior to OTC Restriction	3.25	3.25	3.25	3.38	1.10	1.10	1.10	1.15
Observations	3825	3825	3825	3675	3825	3825	3825	3675
Number of States	51	51	51	49	51	51	51	49
Linear State Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

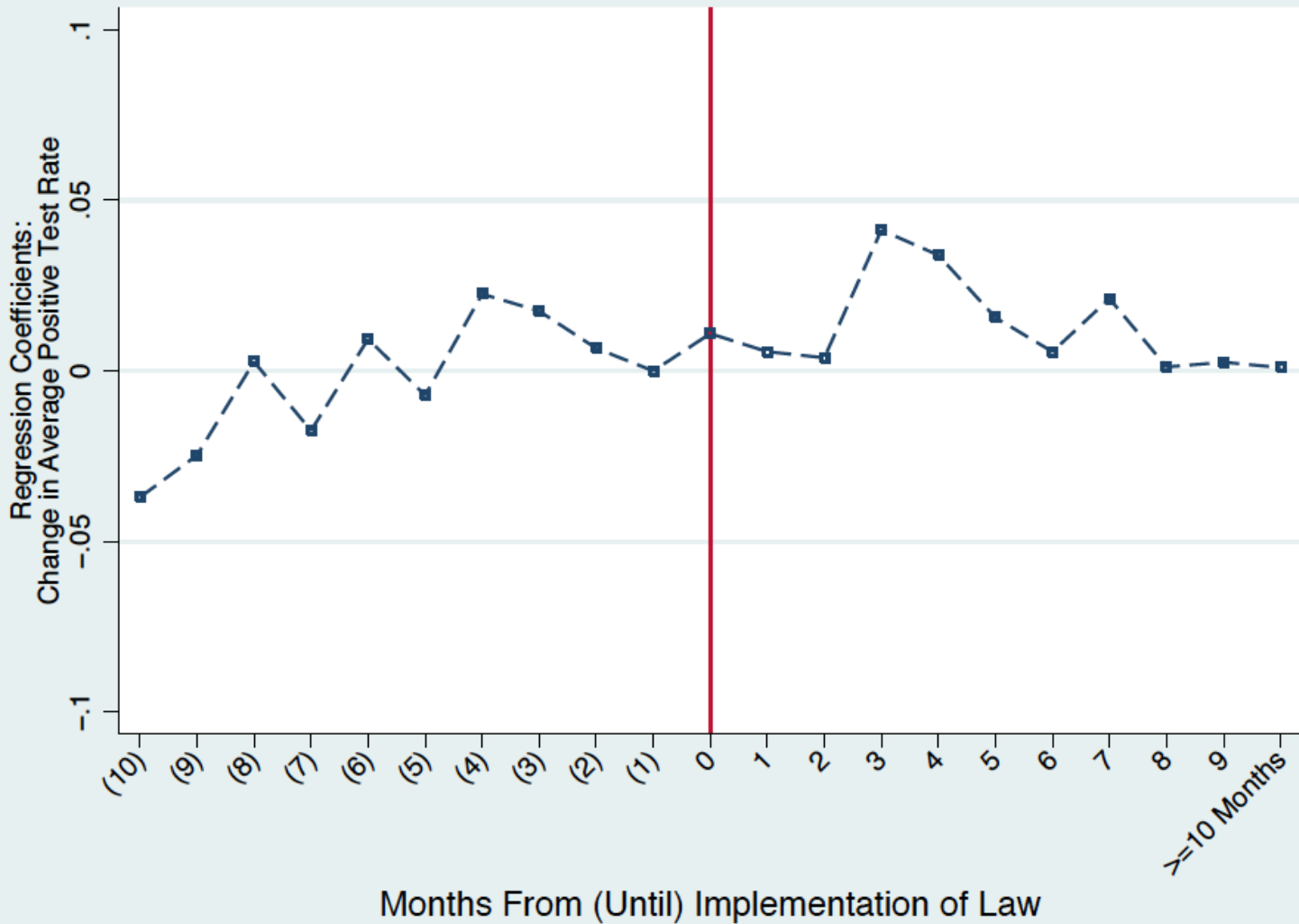
Methamphetamine Market



Methamphetamine Market

	<u>Price per Gram</u> <u>Methamphetamine</u>				<u>Purity</u> <u>Methamphetamine</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	2.89	2.19	-1.83	-2.12	6.15	4.49	3.81	4.18
	(2.94)	(2.93)	(3.54)	(3.57)	(2.58)	(2.55)	(2.24)	(2.26)
Mean Prior to OTC Restriction	49.13	49.13	49.13	48.60	56.64	56.64	56.64	56.32
Observations	2,074	2,074	2,074	1,995	2,074	2,074	2,074	1,995
Number of States	49	49	49	47	49	49	49	47
	<u>Price per Gram</u> <u>Cocaine</u>				<u>Purity</u> <u>Cocaine</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	1.43	1.46	2.18	2.08	0.023	0.020	0.14	0.082
	(2.37)	(2.38)	(2.58)	(2.64)	(1.29)	(1.52)	(1.65)	(1.70)
Mean Prior to OTC Restriction	40.66	40.66	40.66	61.26	40.64	40.64	40.64	61.30
Observations	2,449	2,449	2,449	2,423	2,449	2,449	2,449	2,423
Number of States	50	50	50	48	50	50	50	48
	<u>Price per Gram</u> <u>Crack</u>				<u>Purity</u> <u>Crack</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-3.94	-3.19	-1.39	-1.52	0.19	0.19	-0.42	-0.50
	(2.68)	(2.33)	(3.36)	(3.21)	(1.36)	(1.28)	(1.14)	(1.18)
Mean Prior to OTC Restriction	62.69	62.69	62.69	62.68	63.55	63.55	63.55	63.19
Observations	2,381	2,381	2,381	2,342	2,381	2,381	2,381	2,342
Number of States	50	50	50	48	50	50	50	48
	<u>Price per Gram</u> <u>Heroin</u>				<u>Purity</u> <u>Heroin</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-6.41	-6.44	-4.95	-3.81	3.51	2.40	3.00	3.37
	(12.2)	(11.5)	(11.1)	(11.2)	(2.65)	(2.61)	(3.05)	(3.02)
Mean Prior to OTC Restriction	122.04	122.04	122.04	122.03	45.79	45.79	45.79	45.77
Observations	1,166	1,166	1,166	1,165	1,166	1,166	1,166	1,165
Number of States	44	44	44	43	44	44	44	43
Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

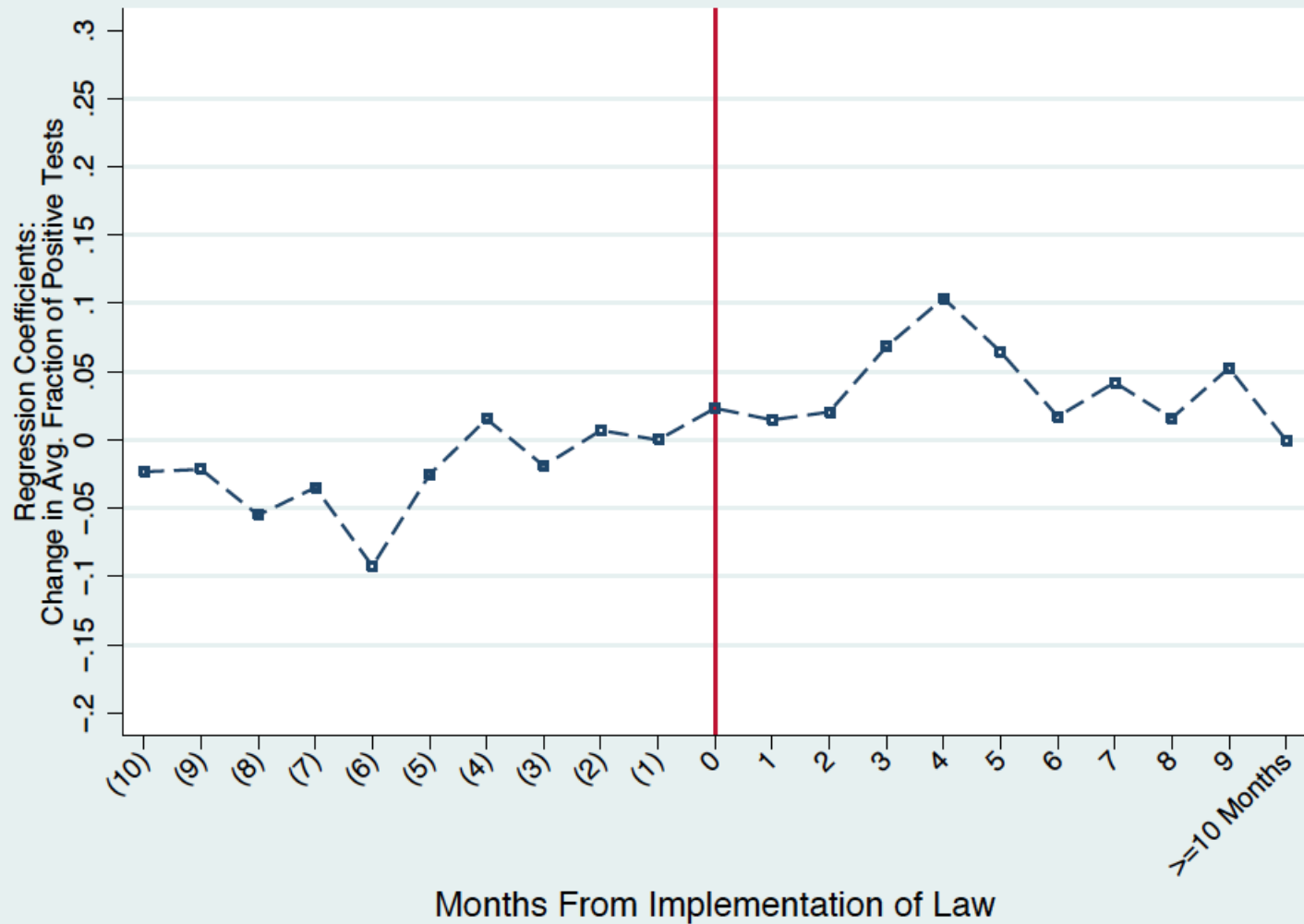
Methamphetamine Use



Methamphetamine Use

	<u>Percent of Tests Positive</u>			
	(1)	(2)	(3)	(4)
OTC Restriction	0.031 (0.018)	-0.012 -0.017	-0.0045 (0.018)	-0.0072 (0.018)
Mean Prior to OTC Restriction	0.365	0.365	0.365	0.362
Observations	3,876	3,876	3,876	3,724
Number of States	51	51	51	49
Linear State Time Trends	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes
Covariates	No	No	No	Yes

Methamphetamine Use



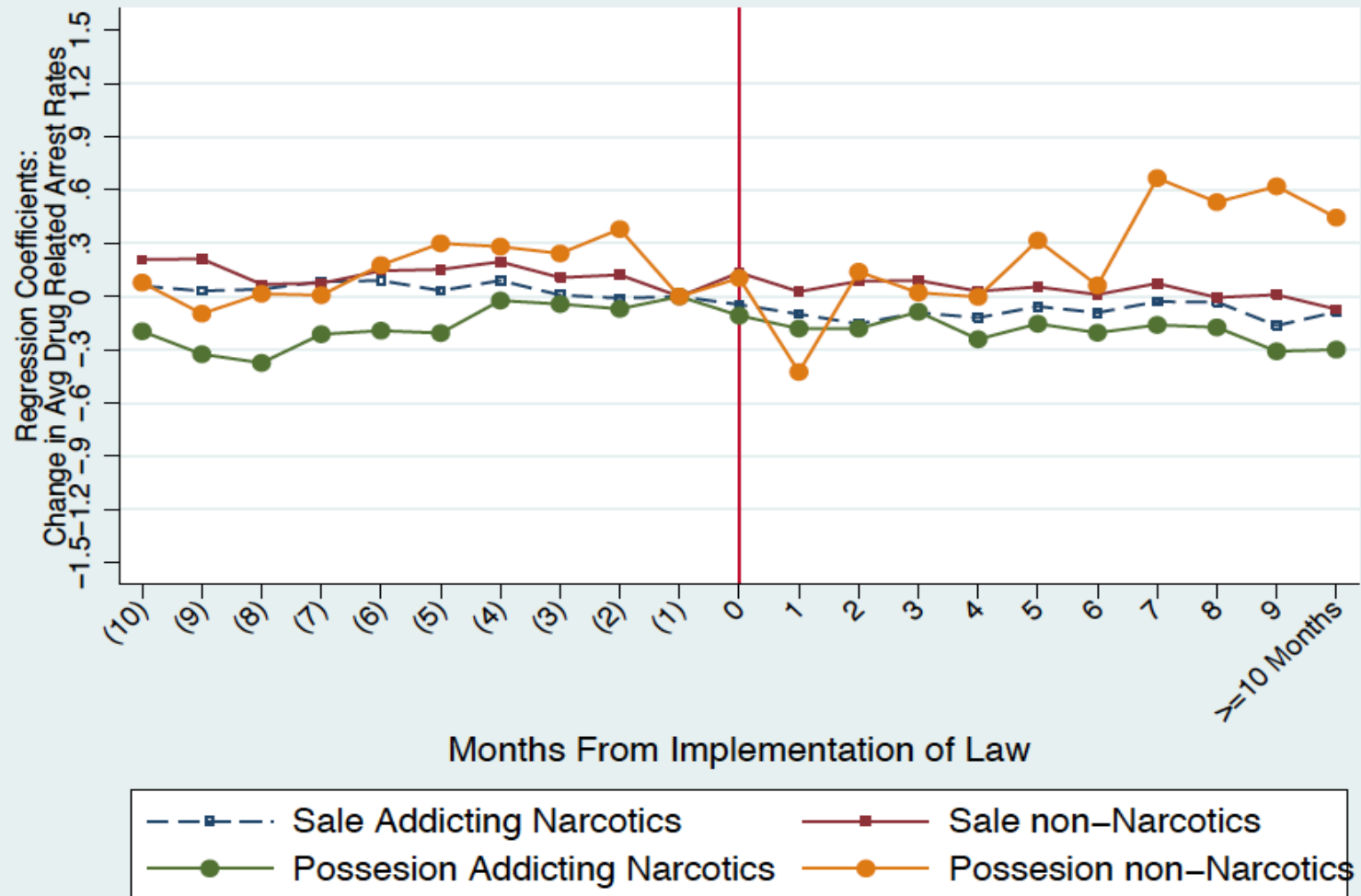
Methamphetamine Use

		<u>Amphetamine</u>				<u>Opioids</u>		
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	0.11 (0.093)	0.083 (0.091)	0.060 (0.057)	0.074 (0.055)	-0.014 (0.15)	-0.093 (0.13)	-0.077 (0.12)	-0.11 (0.15)
Mean Prior to OTC Restriction	0.671	0.671	0.671	0.656	1.912	1.912	1.912	1.928
Observations	2316	2316	2316	2244	2316	2316	2316	2244
Number of States	33	33	33	32	33	33	33	32

		<u>Cocaine</u>				<u>Marijuana</u>		
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-0.33 (0.20)	-0.29 (0.19)	-0.18 (0.16)	-0.22 (0.18)	-0.035 (0.13)	0.0041 (0.12)	0.11 (0.13)	0.11 (0.14)
Mean Prior to OTC Restriction	2.201	2.201	2.201	2.201	1.895	1.895	1.895	1.896
Observations	2316	2316	2316	2244	2316	2316	2316	2244
Number of States	33	33	33	32	33	33	33	32

Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	Yes	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Drug Related Crime



Drug Related Crime

	<u>Sales of Addicting Narcotics</u>				<u>Sales of Dangerous non-Narcotics</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	-0.053 (0.057)	-0.061 (0.042)	-0.13 (0.042)	-0.13 (0.043)	0.012 (0.073)	0.026 (0.071)	-0.087 (0.063)	-0.091 (0.067)
Rate per 10,000	0.726	0.726	0.726	0.728	1.746	1.746	1.746	1.577
Observations	3,477	3,477	3,477	3,327	3,477	3,477	3,477	3,327
Number of States	48	48	48	46	48	48	48	46

	<u>Possession of Addicting Narcotics</u>				<u>Possession of Dangerous non-Narcotics</u>			
	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
OTC Restriction	0.17 (0.11)	0.16 (0.096)	-0.038 (0.091)	-0.025 (0.10)	-0.11 (0.74)	-0.12 (0.76)	-0.24 (0.37)	-0.23 (0.35)
Rate per 10,000	1.572	1.572	1.572	1.577	8.409	8.409	8.409	8.428
Observations	3,477	3,477	3,477	3,327	3,477	3,477	3,477	3,327
Number of States	48	48	48	46	48	48	48	46
Linear State Time Trends	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Quadratic State Time Trends	No	No	Yes	Yes	No	No	Yes	Yes
Covariates	No	No	No	Yes	No	No	No	Yes

Conclusion

- The only significant effect is on the number of lab seizure
- The OTC restriction on ephedrine or pseudoephedrine is costly since it may cause the inconvenience of patients who need cold medicine
- The overall effects of this policy are quite sad

Critique

- Methamphetamine Production

$$\frac{D_{post} - D_{pre}}{D_{pre}} = \frac{p_{post} - p_{pre}}{p_{post}} \left(1 + \frac{LabS_{post} - LabS_{pre}}{LabS_{pre}} \right) + \frac{LabS_{post} - LabS_{pre}}{LabS_{pre}}$$

p_{post} : Lab seizure probability after OTC law

p_{pre} : Lab seizure probability before OTC law

$p_{post} = p_{pre}$: Unbiased

$p_{post} > p_{pre}$: Underestimate production reduction

$p_{post} < p_{pre}$: Overestimate production reduction

Critique

- This paper's story
 - $p_{\text{post}} > p_{\text{pre}}$
 - The police can detect labs by tracing people who are buying ephedrine or pseudoephedrine
 - Increase p_{post}
- Our story
 - $p_{\text{pre}} > p_{\text{post}}$
 - Knowing the OTC law has been passed, drug suppliers will never buy any ephedrine or pseudoephedrine in the drug stores
 - Drug dealers may act more carefully
 - Decrease p_{post}

Critique

- $p_{\text{post}} > p_{\text{pre}}$
 - Underestimate production reduction
 - Significant reduction in lab seizure number can lead to significant reduction in production
- $p_{\text{pre}} > p_{\text{post}}$
 - Overestimate production reduction
 - Significant reduction in lab seizure number leads to nothing

Critique

- Decrease in small labs may not significantly affect the whole production

Critique

	<u>Number of Labs Seized</u>				<u>Lab Capacity Under 2 Ounces</u>			
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Covariates	No	No	No	Yes	No	No	No	Yes

Critique

- Methamphetamine Market
- Drug dealers who sell drugs to the police
 - Are too young, too simple and sometimes naive
 - May act irrationally
 - The market should be conducted by sophisticated drug dealers
- Selection bias

Critique

- Drug Related Crime
- Arrest data does not distinguish methamphetamine
 - Narcotic or non-narcotic
 - Addicting or dangerous
- We do not know the proportion of methamphetamine related crime in this data

Thanks